1. HTML/CSS Revision

1. Write HTML, CSS3 code for following animation effects (A)
a. changing background color of div from red to yellow.
b. moving div from Left to right, right to bottom, bottom to left and left to top.
c. put delay of 2s in animation.
d. Change animation-iteration count property of program 2 to run infinite times
e. Change animation direction property to run it in reverse order.
f. Change value of animation-timing-function property to change the speed of animation
2. Write an HTML snippet with CSS for following effects: (A)
a. Link after visited red color before visited orange color
b. Add special style(change color,font size etc) to first line of paragraph
c. Add special style (change color,font size etc) to first letter of paragraph
d. Change background color to yellow when user select particular text
e. Add special style to an element when user mouse over it.
3. Design complete personal website using CSS. (B)

2. Bootstrap Revision

Desing clone of darshan university home page using bootstrap. (A)

3. Demonstrate the use of popup boxes in JavaScript

1. WAP in JavaScript to make simple calculator using popup box. (A)
2. WAP in JavaScript to check whether the given no. is prime or not. (A)
3. WAP in JavaScript to ﬁnd the factorial of given number. (A)
4. WAP in JavaScript to print the Fibonacci series of a number. (A)
5. WAP in JavaScript to check whether the given number is palindrome or not. (A)
6. WAP in JavaScript to check whether the given number is Armstrong or not. (A)

4. Implement basic JavaScript programs

1. WAP in JavaScript to print the factors of given number. (A)
2. WAP in JavaScript to prime number between the given range of number. (A)
3. WAP to display given patterns using JavaScript. (A)
4. WAP in JavaScript to print the GCD of two number. (A)
5. Write a JavaScript to take 2-digit number and then separate these 2 digits, then multiply ﬁrst digit by itself for second digit times. (B)
(For example, 23 should be separated as 2 and 3. 2 should multiply with itself 3 times).
6. Write an HTML page with JavaScript that takes a number from popup box in the range 0-999 and display it in words. If the number is out of range, it should show “out of range” and if it is not a number ,it should show “not a number” message in the result box. (A)
7. Write an HTML ﬁle with JavaScript that ﬁnds position of ﬁrst occurrence of vowel “a”, last occurrence of vowel “a” in a given word and returns the string between them. For example, ajanta-

then script would return ﬁrst occurrence of “a”-that is position 1 and last
occurrence-6 and string between them is “jant”. (B)

5. Implement basic array programs in JavaScript

1. Demonstrate the use of array to ﬁnd maximum number. (A)
2. WAP to read an array from the user and sort them in ascending order. (A)
3. WAP to read an array from the user and sort them using bubble sort. (A)
4. WAP to read an array form the user and sort them using quick sort. (B)
5. Write a JavaScript that uses a loop that searches a word in sentence held in an array, returning the index of the word. (B)

6. Render Array elements using JavaScript

1. WAP to display Faculties stored in array. (A)
2. Demonstrate different array functions like push, pop, shift, unshift, splice, sort etc… (A)
3. WAP to display students stored in array. (A)
4. WAP to display products stored in array. (B)

7. Implement Validation of form ﬁelds in JavaScript

1. WAP to create a basic graphical Calculator using HTML and JavaScript. (A)
2. WAP to create a static login page using HTML and JavaScript. (B)
3. WAP to create a scientiﬁc graphical calculator using HTML and JavaScript. (C)
4. Write a JavaScript program to validate user data given from the HTML form (A)

1) username must be of minimum 8 characters
2) password must contain at least 1 digit and 1 special character and should be between 8-12 characters
3) password and repeat password must be same
4) age must be greater than 18 (calculate from date of birth)
5) enrollment must be of 11 digits
6) email validation

8. Demonstrate Document Object Model methods in JavaScript

1. WAP to change background color on click of button. (A)
2. WAP to recognize which mouse event is ﬁred. (A)
3. WAP to recognize which keyboard event is ﬁred. (B)
4. WAP to recognize which form event is ﬁred. (B)
5. WAP to demonstrate change in properties of HTML element using JavaScript. (A)
6. WAP to prepare student registration form and validate it using JavaScript. (A)

7. WAP to perform CRUD operation on Array using HTML and JavaScript. (C)
8. Write a JavaScript that handles following mouse events. Add necessary elements. (B)
If the mouse is over the heading should turn yellow and if the mouse goes out of the heading it should turn black.
If fndtime button is clicked show time and date information.
If button named “red” is clicked, background should change to red and If button named “green” is clicked, background should
change to green.
9. Write a JavaScript that handles following mouse events. Add necessary elements. (B)
JavaScript gives the key code for the key pressed.
If the key pressed is “a”,”e”,”i”,”o”,”u”, the script should announce that vowel is pressed.
When the key is released background should change to blue.

9. Demonstrate ES6 features in JavaScript – 1

1. WAP to demonstrate callbacks in JavaScript. (A)
2. Demonstrate the difference between let and var. (A)
3. Demonstrate the default parameter while using a function. (A)
4. Create a Snake game using Java script. (C)

10. Demonstrate ES6 features in JavaScript – 2

1. Demonstrate the Arrow functions. (A)
2. Write JavaScript that handles following mouse event. (B)
• If mouse left button pressed on browser, it displayed message “Left Clicked”.
• If mouse right button pressed on browser, it displayed message “Right Clicked”.
3. Write a JavaScript having a list of checkbox and by clicking on checkbox, it should show list of selected value in comma separated format. (e.g. list of roll number as checkbox value, and display selected roll number in comma separated format) (C)

11. Demonstrate ES6 features in JavaScript – 3

1. Demonstrate how to create a class in Java Script. (A)
2. Demonstrate the spread operator. (A)
3. Demonstrate the ‘for of’ loop. (A)
4. Demonstrate the Array and Object Destructuring. (B)

12. Perform JQuery Operations

1. write a jQuery to fade object when clicked on it. (A)
2. write a jQuery to slideup, slidedown and slidetoggle image. (A)
3. write a jQuery program to set content of h1 to "Hello World" when use click on button. (A)
4. write a jQUery program to get content of textbox and set to paragraph with "data" class. (B)

5. write a jQuery program to append new list item with "Hello World" to existing unordered list when clicked on button. (A)
6. write a jQuery program to append new row to existing table when clicked on button. (A)
7. write a jQuery program to change background color to be "red" to all the childrens of div with "myDiv"as id when user click on a button. (B)
8. write a jQuery to hide all siblings when clicked on a button. (A)
9. write a jQuery program to hide element with id ""myEle"" when clicked on a button (A)
10. write a jQuery program to hide all elements with class ""myEle"" when clicked on a button. (A)
11. write a jQuery program to toggle hide/show element with id ""myEle"" when clicked on a button (A).
12. write a jQuery program to change background color of all input tags when clicked on a button (A)
13. write a jQuery program to hide an element which was clicked (B).
14. write a jQuery program to set background of an input element to red when in focus and back to white when not in focus. (A)
15. write a jQuery program to set background of an element to red when mouse enter on all paragraphs (A)
16. write a jQuery program to print current mouse coordinates on console. (B)
17. write a jQuery program to validate the darshan enrollment no when user leave the textbox (blur). (B)
18. write a jQuery program to validate complete student registration form. (C)

13. Perform CRUD operation on a static array using jQuery

Perform CRUD operation on a static array with FirstName, LastName, Age, Semester, RollNumber, ImagePath etc.. Fields using jQuery

14. Setting up ReactJS development Environment

1. Setting up react environment. (A)
2. Hello world webapp using ReactJS. (A)
3. Demonstrate the use of JSX. (A)
4. WAP to create a simple class component in ReactJS. (A)
5. WAP to create a simple function component in ReactJS. (A)

15. Create Hello World ReactJS app

1. Create a function component in separate file and link with App.js (A)
2. Create a class component in separate file and link with App.js (B)

16. Demonstrate props in ReactJS

1. Demonstrate the ReactJS props. (A)
2. Demonstrate the Event Handling in ReactJS. (A)
3. WAP in ReactJS to display the element if it has attribute called isDisplay to be true (using conditional rendering) (A)

17. Demonstrate the use of map method in ReactJS

1. Demonstrate the use of map method in ReactJS to display array. (A)
2. Display Faculties stored in array using ReactJS. (B)
3. Display Students stored in array using ReactJS. (B)
4. Display Products stored in array using ReactJS (C)

18. Demonstrate Property Drilling in ReactJS

1. Create a react application with following components.
- create a component named "F" which print one state value named "name" from "App" Component.
- create component named "E" which contains "F" component.
- create component named "D" which contains "E" component.
- create component named "C" which contains "D" component.
- create component named "B" which contains "C" component with textbox and a button and when button is clicked set the state value named "name" from "App" component.
- create component named "A" which contains "B" component.
- "App" component should contains "A" component. (A)

2. Create a react application with following components.
- create a component named "F" which print one state value named "name" from "App" Component.
- create component named "E" which contains "F" component.
- create component named "D" which contains "E" component.
- create component named "C" which contains "D" component.
- create component named "B" which contains "C" component with button and when button is clicked set the state value named "name" from "App" component with the value of textbox from component "A".
- create component named "A" which contains "B" component and a textbox.
- "App" component should contains "A" component. (B)

3. Implement static login implementation with property drilling. (C)

19. Perform Static Authentication and manage the state across layout using property drilling

Create an app using ReactJS with login functionality and use the current user state throughout the application using property drilling.

20. Demonstrate Routing in ReactJS

1. Implement Routing in ReactJS. (A)
2. Develop basic website using 5 different component (pages) and implement Routing in it. (i.e. About, Contact etc…) (A)
3. Develop full static website using 15 different component (pages) and implement Routing in it. (i.e. About, Contact etc…) (C)

21. Demonstrate the use of hooks in ReactJS

1. Demonstrate useState hook in ReactJS. (A)
2. Demonstrate useEffect hook in ReactJS (A)

22. Create GUI Calculator using ReactJS

1. WAP to create a simple calculator using ReactJS. (A)
2. WAP to create a scientific calculator using ReactJS. (C)

23. Implement CRUD operation on Array in ReactJS

1. WAP to do CRUD operation on products stored as array using ReactJS. (A)
2. WAP to do CRUD operation on students stored as array using ReactJS. (B)
3. WAP to do CRUD operation on faculties stored as array using ReactJS. (C)

24. Perform CRUD operation on MockAPI using ReactJS

1. Create a MockAPI online with following fields. (A)
- FacultyID
- FacultyName
- FacultyExp

• FacultyImage
2. Perform CRUD operation on MockAPI using ReactJS. (minimum 3 mock api) (A)
3. Perform CRUD operation on MockAPI using ReactJS. (minimum 8 mock api) (B)
4. Perform CRUD operation on MockAPI using ReactJS. (minimum 15 mock api) (C)

25. Perform Static Authentication and manage the state acrose layout using useContext

Create an app using ReactJS with login functionality and use the current user state throughout the application using useContext hook.

26. Create a mini project for library management system

27. Create a mini project for library management system

28. Create a mini project for attendance management system

29. Create a mini project for attendance management system

30. Create a mini project for attendance management system